SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone

Ijaz Ahmad

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GitHub URL:

https://github.com/Ijaz-Ahmad7/spacex-falcon9-capstone

Executive Summary

- Goal: Predict whether the Falcon 9 first stage will land successfully, to estimate launch cost.
- Data: 90 Falcon 9 launches collected via SpaceX REST API and Wikipedia web scraping.
- Methods: Data wrangling → EDA → SQL analysis → interactive map + dashboard → 4 ML classifiers
- Key Results:
  - Landing success improved from 0% (2010–2013) to ~85% (2019–2020).
  - Best model: Logistic Regression with 83.3% test accuracy.
  - Success rate varies by orbit — GTO lowest at 51.9%.
- Impact: Model can support launch cost estimation and bid decisions.

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Introduction: Background & Problem

- SpaceX's cost advantage: Falcon 9 launches cost $62M vs. competitors' $165M+.
- Why? Reusable first stage — landing the booster saves the cost of building a new one.
- Key question: Can we predict whether the first stage will land successfully?
- Business value: If yes, we can estimate launch cost and help companies bid against SpaceX.
- Target variable: Class — 1 = successful landing, 0 = failed landing.

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Data Collection — SpaceX REST API

- Endpoint: GET https://api.spacexdata.com/v4/launches/past
- Method: requests.get() → response.json() → pd.json_normalize() → DataFrame
- Extracted fields: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial
- Result: 107 launches collected into dataset_part_1.csv
- Fallback: Used IBM's static JSON snapshot because the live API rate-limits Colab requests

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Data Collection — Web Scraping (Wikipedia)

- Source: Wikipedia — List of Falcon 9 and Falcon Heavy launches
- Tools: requests.get() + BeautifulSoup(html, 'html.parser')
- Challenge: Wikipedia returned HTTP 403 (bot detection). Fixed by adding a User-Agent header.
- Process:
  - Fetch page HTML → parse with BeautifulSoup
  - Find launch table → extract rows → build DataFrame
  - Alternative: pd.read_html() handles rowspans automatically
- Result: Launch records with Payload Mass, Orbit, Launch Outcome, Booster Landing — merged into dataset_part_2.csv (90 rows × 18 columns)

GitHub URL:

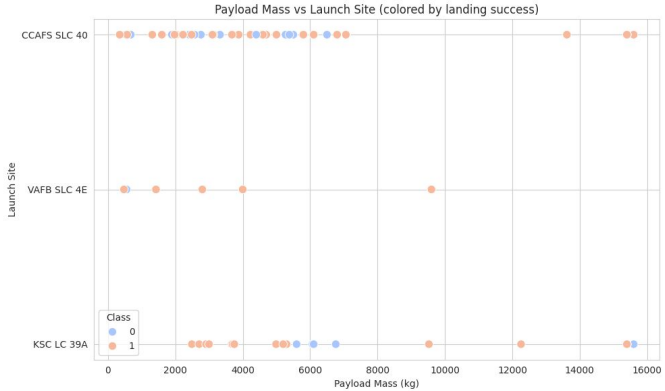
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Data Wrangling Methodology

- Merge: Combined API data (107 rows) and scraped Wikipedia data into a single dataset
- Missing values: Checked and handled NaN in payload mass, landing pad, and outcome columns
- Type conversion: Converted Date to datetime; cast booleans for GridFins, Reused, Legs
- Target variable: Created Class — 1 = successful landing, 0 = failure
- Feature engineering: Added Longitude and Latitude for launch sites (used for Folium map)
- Final dataset: 90 rows × 18 columns → saved as dataset_part_2.csv

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Exploratory Data Analysis — Visualizations

- Tools: Matplotlib + Seaborn
- Six charts were created, each answering a specific question:
  - Flight Number vs Launch Site — which site launches most?
  - Payload Mass vs Launch Site — does payload size relate to success?
  - Success Rate by Orbit — which orbits are hardest to land from?
  - Flight Number vs Orbit — how has orbit usage changed over time?
  - Payload Mass vs Orbit — does mass × orbit affect outcome?
  - Yearly Success Trend — how has success improved over the years?
- Key finding: Landing success went from 0% (2010–2013) to ~85% (2019–2020)

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Exploratory Data Analysis — SQL

- Tool: SQLite (in-memory) via Python's sqlite3
- Seven SQL queries were run:
  - Q1: Total payload mass by booster block
  - Q2: Average payload mass by launch site
  - Q3: Number of launches per launch site
  - Q4: Total payload mass by launch site
  - Q5: Successful vs failed landings
  - Q6: Number of launches per year
  - Q7: Max payload mass by orbit
- Key findings:
  - CCAFS SLC 40 is the busiest pad: 55 launches
  - KSC LC 39A carries the heaviest average payload: 7,606 kg
  - Overall landing success rate: 67% (60 success / 30 fail)

GitHub URL:

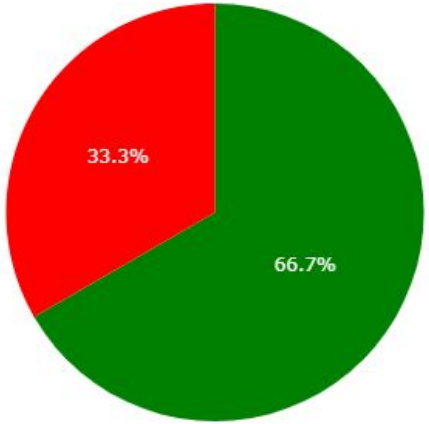
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Interactive Visual Analytics

- Folium Map — Launch Site Locations
  - Shows all 3 launch sites: CCAFS SLC 40, VAFB SLC 4E, KSC LC 39A
  - Markers colored by site; circles show landing success (green) vs failure (red)
  - Saved as spacex_launch_sites.html
- Plotly Dash Dashboard
  - Pie chart: 67% success vs 33% failure
  - Interactive scatter plot: Payload Mass vs Landing Outcome
  - Dropdown filter: view by specific launch site or all sites
  - Dashboard updates live when dropdown changes
- Purpose: Let users explore launch patterns interactively instead of reading static tables

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EDA Results — Chart 1: Flight Number vs Launch Site

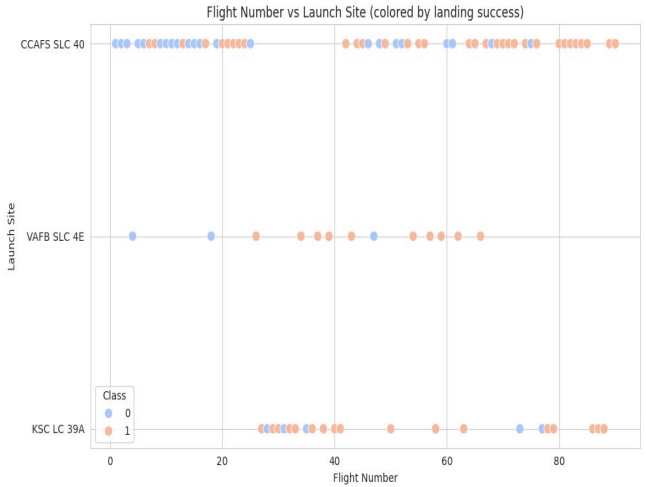
- Chart type: Scatter plot (Seaborn)
- What it shows: Every Falcon 9 launch plotted by flight number (x-axis) and launch site (y-axis)
- Color coding: Blue = Class 0 (failure), Orange = Class 1 (success)

Key observations:

- CCAFS SLC 40 has the most launches (~55)
- Early flights (0–30) are mostly blue (failures)
- Later flights (50+) are almost all orange (successes)
- KSC LC 39A only appears after flight ~20 (opened 2015)

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EDA Results — Chart 2: Payload Mass vs Launch Site

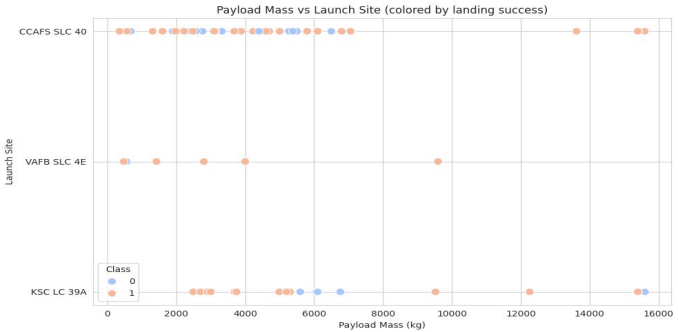
- Chart type: Scatter plot (Seaborn)
- What it shows: Payload mass (kg) vs launch site, colored by landing outcome
- Color coding: Blue = failure, Orange = success

Key observations:

- Payloads under 4,000 kg at CCAFS: many blue (failures)
- Payloads above 8,000 kg: almost all orange (successes)
- VAFB SLC 4E: small number of launches, mixed results
- KSC LC 39A: consistently orange (successes) with heavy payloads

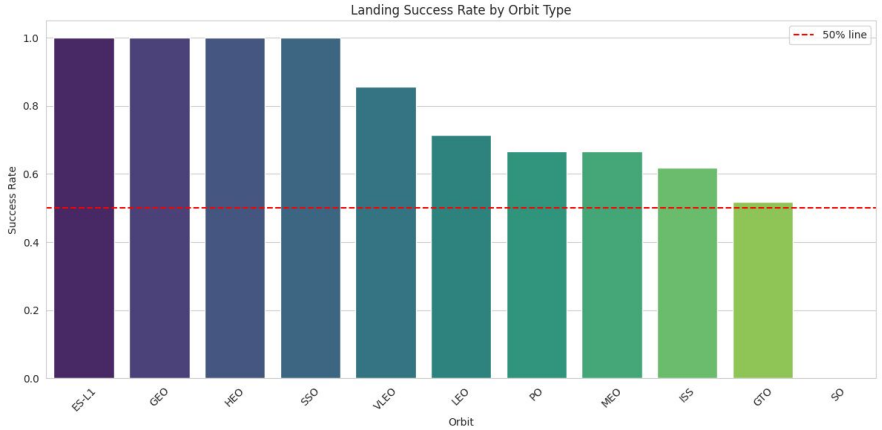
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EDA Results — Chart 3: Success Rate by Orbit

- Chart type: Bar chart (Seaborn)
- What it shows: Landing success rate (%) for each orbit type
- Sorted: highest → lowest success rate



Key observations:

- 100% success: ES-L1, GEO, HEO, SSO (rare orbits, small sample)
- 85.7%: VLEO
- 71.4%: LEO
- 66.7%: PO, MEO
- 61.9%: ISS
- 51.9%: GTO — the hardest common orbit
- 0%: SO (single launch, failed)

Insight: GTO missions carry heavy payloads to high orbits, leaving less fuel for the booster to land — that's why success is lowest there.

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EDA Results — Chart 4: Flight Number vs Orbit Type

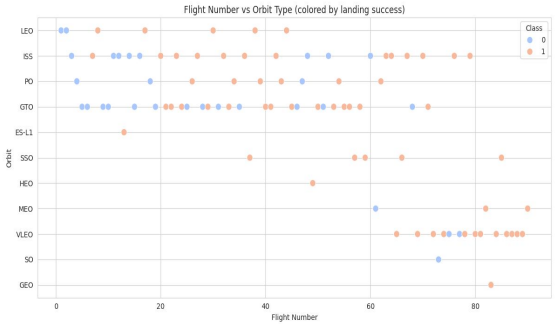
- Chart type: Scatter plot (Seaborn)
- What it shows: Flight number (x-axis) vs orbit type (y-axis), colored by landing outcome
- Color coding: Blue = failure, Orange = success

Key observations:

- GTO and ISS dominate — the most common orbits
- Early flights (1–30): many blue dots across all orbits
- Later flights (50+): almost all orange
- Rare orbits (ES-L1, SSO, HEO, MEO, VLEO, SO, GEO) appear only in later flights
- Clear transition from blue → orange as flight number increases

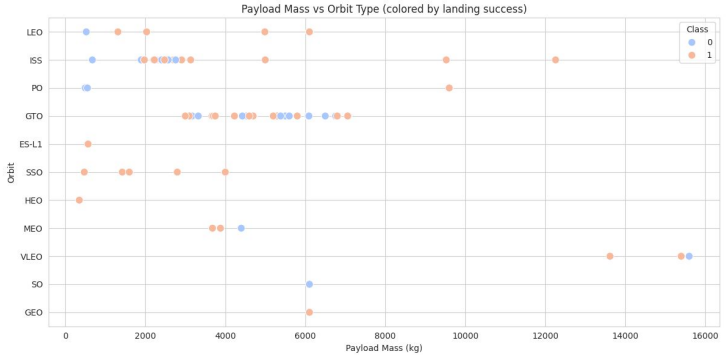
GitHub URL:

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EDA Results — Chart 5: Payload Mass vs Orbit Type

- Chart type: Scatter plot (Seaborn)
- Shows: Payload mass (kg) vs orbit type, colored by landing outcome
- LEO: spans light (500 kg) to heavy (15,600 kg) payloads
- GTO: clustered around 3,000 - 6,000 kg
- Rare orbits (ES-L1, HEO, SSO): only light payloads
- Takeaway: No simple "heavier = fail" rule payload mass interacts with orbit, and GTO is hardest despite moderate payloads



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EDA Results — Chart 6: Yearly Success Trend

- Chart type: Line chart (Seaborn)
- Shows: Landing success rate per year, 2010–2020

Year-by-year:

- 2010–2013: 0% (all failures)
- 2014–2015: 33%
- 2016: 63%
- 2017: 83%
- 2018: 61% (dip — a few failures)
- 2019: 90%
- 2020: 84%



Takeaway: SpaceX went from zero successful landings to ~85% success in under a decade — the strongest evidence of program maturity.

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SQL Results — Launch Site Analysis

Query 1 — Total payload mass by launch site:

- CCAFS SLC 40: 305,151 kg
- KSC LC 39A: 167,342 kg
- VAFB SLC 4E: 76,953 kg

Query 2 — Number of launches per site:

- CCAFS SLC 40: 55 launches
- KSC LC 39A: 22 launches
- VAFB SLC 4E: 13 launches

Takeaway: CCAFS SLC 40 is the busiest pad by far — both in launch count and total payload lifted.

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SQL Results — Payload Analysis

Query 1 — Total payload mass by booster block:

- Block 5: 343,646 kg (39 launches)
- Block 3: 79,813 kg (15 launches)
- Block 4: 53,901 kg (11 launches)
- Block 1: 48,998 kg (19 launches)
- Block 2: 23,089 kg (6 launches)

Query 2 — Max payload mass by orbit:

- VLEO: 15,600 kg
- ISS: 12,259 kg
- PO: 9,600 kg
- GTO: 7,076 kg
- LEO / SO / GEO: ~6,105 kg

Takeaway: Block 5 is the modern workhorse — most launches and most total payload lifted. VLEO carries the heaviest single payload.

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SQL Results — Success Rates & Rankings

Query — Successful vs failed landings:

- Successful (Class 1): 60 launches
- Failed (Class 0): 30 launches
- Overall success rate: 67%

Ranking — Top orbits by success rate:

1. ES-L1 / GEO / HEO / SSO — 100%
2. VLEO — 85.7%
3. LEO — 71.4%
4. PO / MEO — 66.7%
5. ISS — 61.9%
6. GTO — 51.9%
7. SO — 0%

Takeaway: Overall success sits at 67%, but high-payload GTO missions drag the average down. Light-payload missions to high orbits rarely fail.

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SQL Results — Time Analysis

- 2010–2013: 1, 0, 1, 3 launches
- 2014–2016: 6, 6, 8 launches
- 2017–2020: 18, 18, 10, 19 launches
- Trend: steady growth, 2020 was the busiest year

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SQL Results — Average Payload by Site

Query — Average payload mass by launch site:

- KSC LC 39A: 7,606 kg
- VAFB SLC 4E: 5,919 kg
- CCAFS SLC 40: 5,548 kg

Takeaway: KSC LC 39A handles the heaviest average payloads — it's the pad used for the most demanding missions (heavy GTO satellites, deep-space payloads).

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Interactive Map — Launch Sites & Landing Outcomes

- Tool: Folium (Python)
- Base map: Esri World Light Gray (clean, no API key needed)
- Three launch sites marked:
  - CCAFS SLC 40 (blue marker) — Cape Canaveral, Florida
  - KSC LC 39A (red marker) — Kennedy Space Center, Florida
  - VAFB SLC 4E (green marker) — Vandenberg, California
- Landing outcome circles:
  - Green circle = successful landing
  - Red circle = failed landing
- Density: 60 green circles + 30 red circles clustered around the sites
- Saved as: spacex_launch_sites.html (interactive — open in browser to zoom/pan)



Proximity analysis:

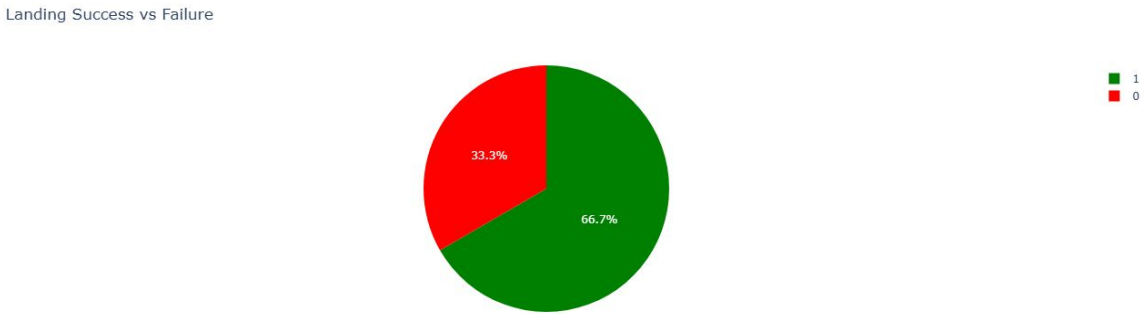
- CCAFS SLC 40 and KSC LC 39A are ~30 km apart on Florida's east coast
- VAFB SLC 4E is isolated on California's central coast

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Plotly Dash — Interactive Dashboard

- Tool: Plotly Dash (running inline in Colab)
- Pie chart: Landing success vs failure
  - Success (Class 1): 66.7% (green)
  - Failure (Class 0): 33.3% (red)
- Scatter plot: Payload Mass (x-axis) vs Landing Outcome (y-axis)
  - Dropdown filter: select "All Sites" or a specific launch site
  - Updates live when you change the selection
- Purpose: Let users interactively explore the data — no need to read static tables
- Saved as: dashboard runs in the notebook via app.run(jupyter_mode='inline')



Predictive Analysis — Models & Results

- 4 models trained with GridSearchCV (10-fold cross-validation):
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Decision Tree
  - K-Nearest Neighbors (KNN)
- Test Accuracy:
  - Logistic Regression — 83.3%
  - SVM — 83.3%
  - KNN — 83.3%
  - Decision Tree — 77.8%
- Best model: Logistic Regression
  - Chosen for top accuracy + interpretability
  - Best params: C=0.1, penalty='l2', solver='lbfgs'

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Predictive Analysis — Conclusion

		Predicted	
		Fail	Success
Actual	Fail	3	3
	Succ	0	12

What it means:

- 12/12 successes correctly predicted ✅
- 3/6 failures caught — misses half ⚠️
- Model is reliable for successes, conservative on failures

Conclusion:

- Landing success improved from 0% (2010) → ~85% (2020)
- Model can support launch cost estimation — a successful landing saves ~$50M per launch
- Business value: Helps companies estimate SpaceX launch costs and bid competitively

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